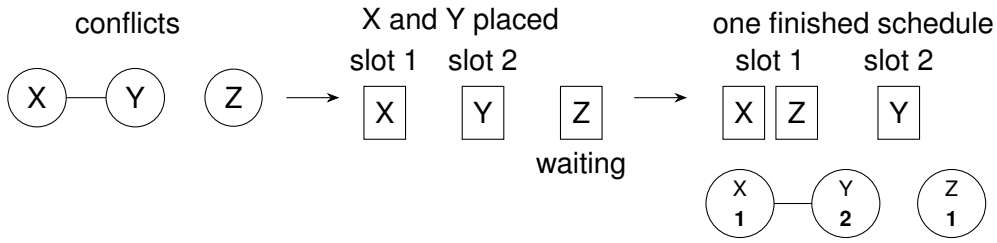
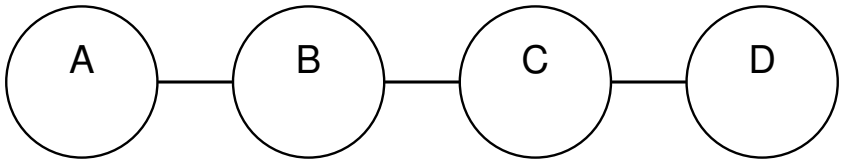
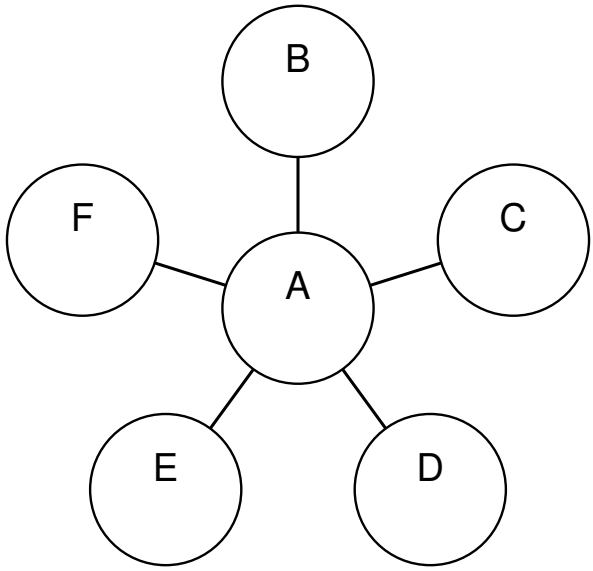
Each circle is one activity card. A line means its two cards cannot share a slot. Put each card in exactly one slot. A slot can hold any number of cards if none of them conflict. Only marked circles are activities; line crossings add none. Write a card's slot number in its circle.



Problem 1: Use your cards to schedule each network with the fewest slots you can.

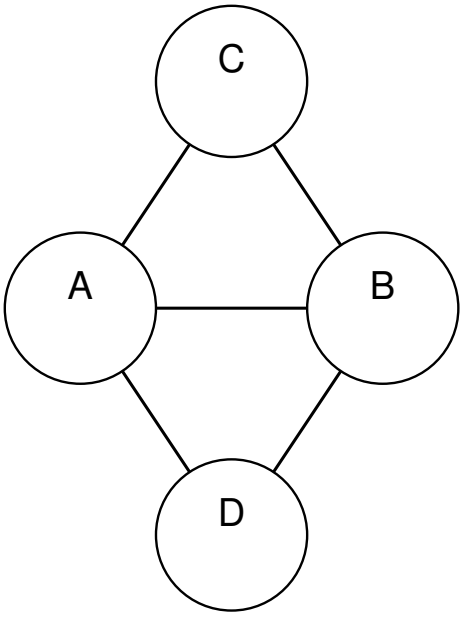


Fewest slots: _____

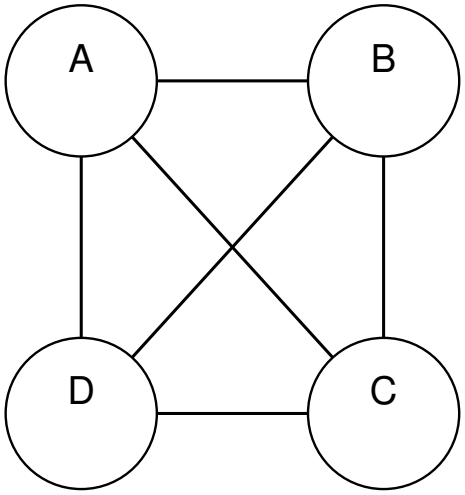


Fewest slots: _____

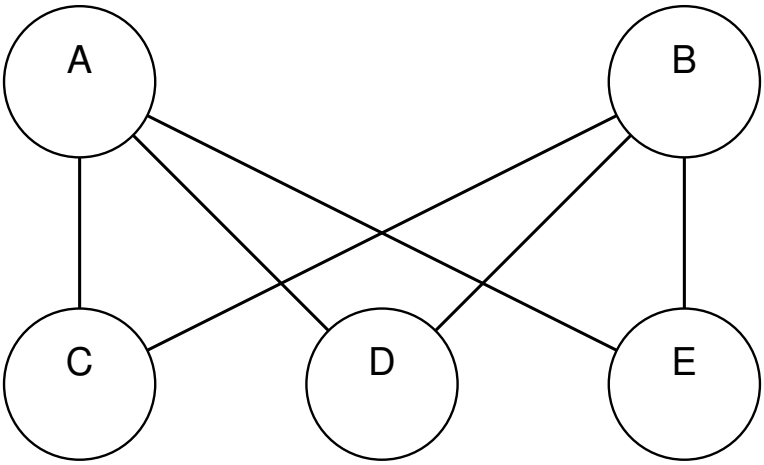
Problem 2: Find the fewest slots for each network. Show why fewer slots cannot work.



Fewest slots: _____

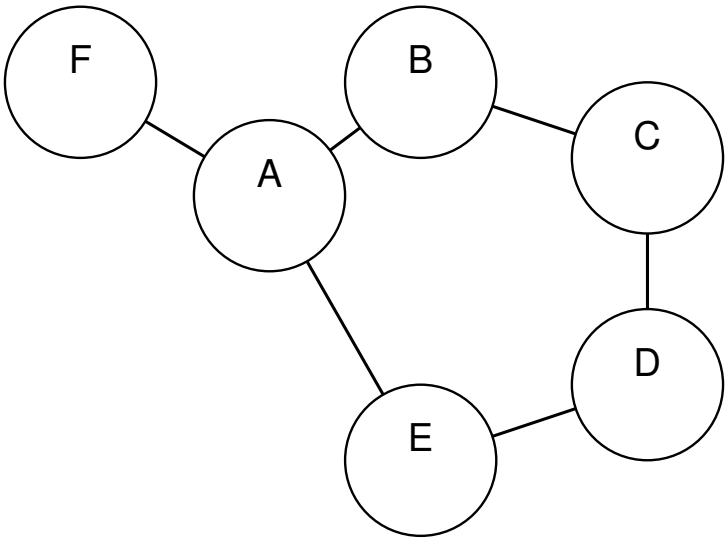


Fewest slots: _____

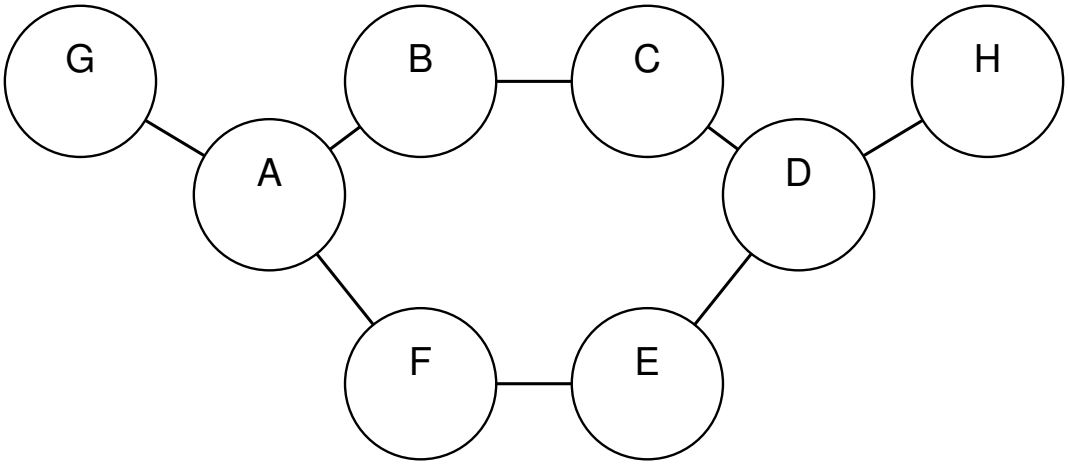


Fewest slots: _____

Problem 3: Find the fewest slots for each network. Also find the largest group of cards in which every pair conflicts. Does that group's size always give the fewest slots?

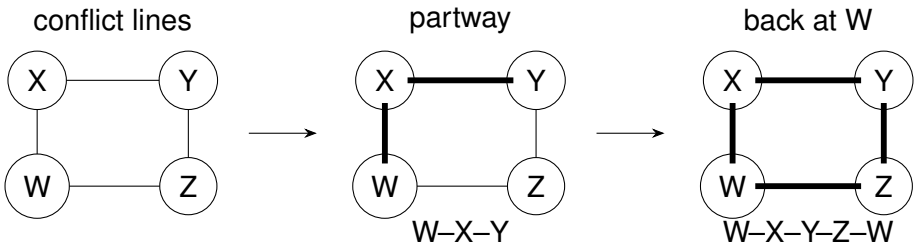


Fewest slots: _____ Largest group: _____

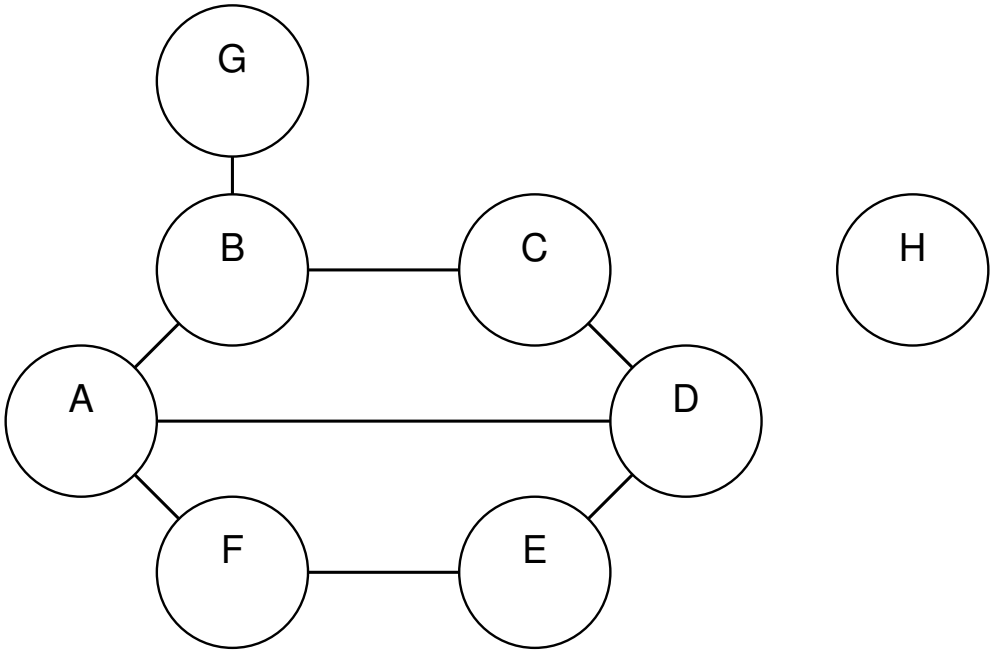


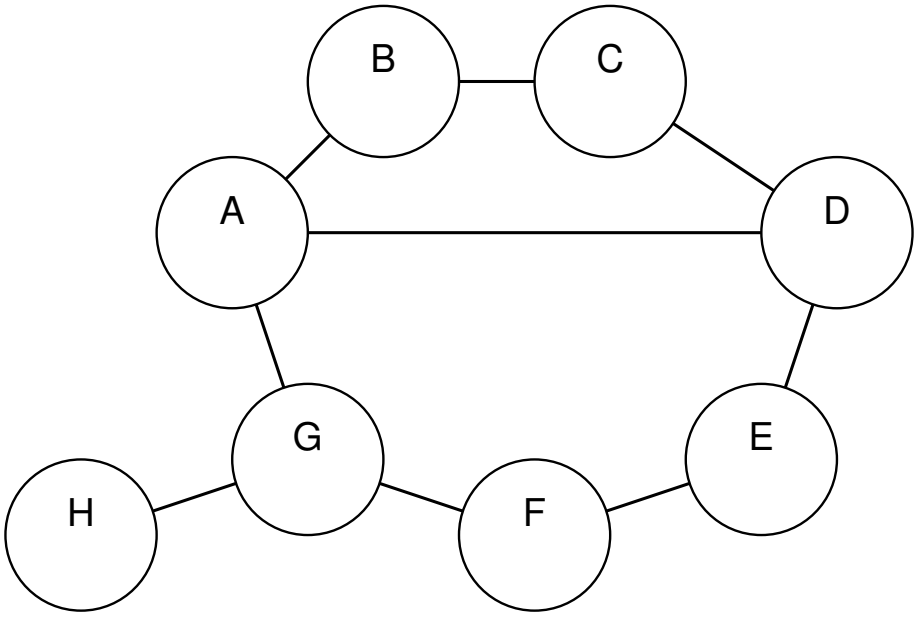
Fewest slots: _____ Largest group: _____

A cycle follows conflict lines through at least three different circles and back to its start. Only the start repeats, at the end.

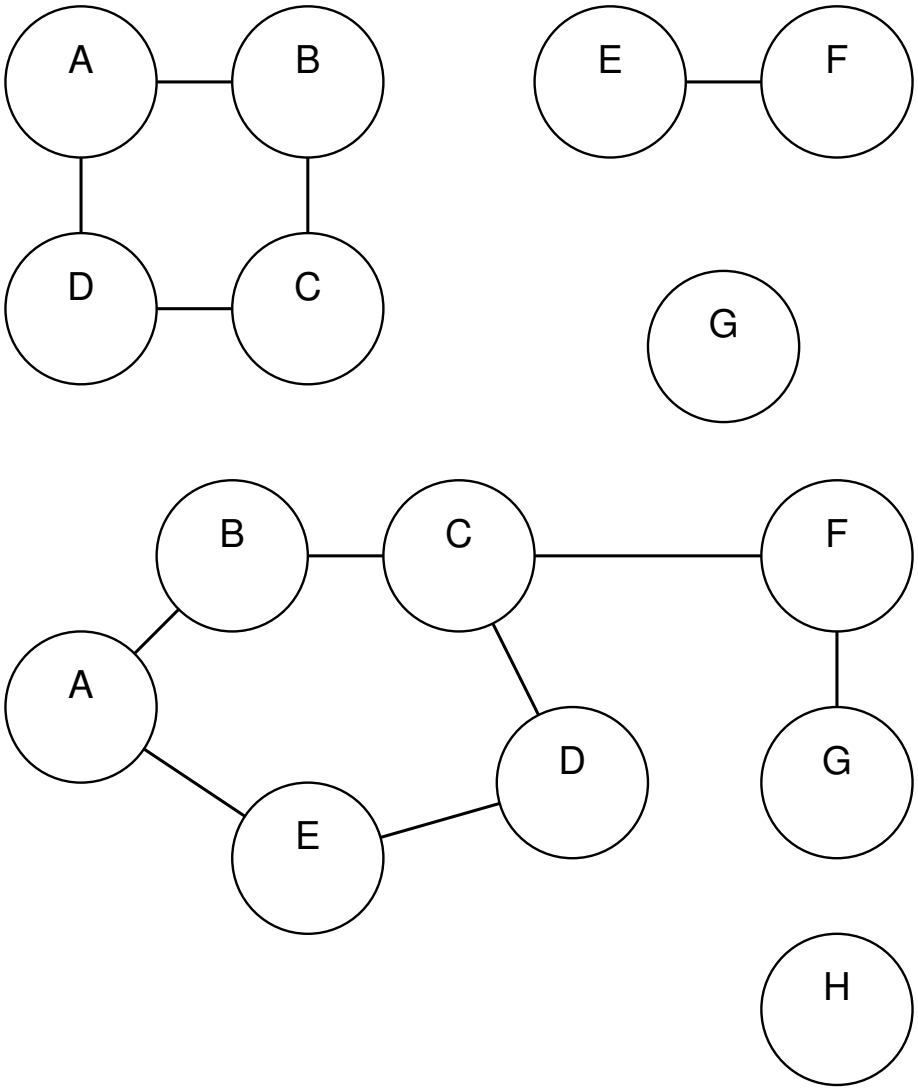


Problem 4: For each network, show a two-slot schedule if one exists. If it cannot fit in two slots, mark a cycle that explains why.





Problem 5: Make a rule that decides which networks fit in two slots. Use it on these networks. Explain why your rule works for networks of any size.

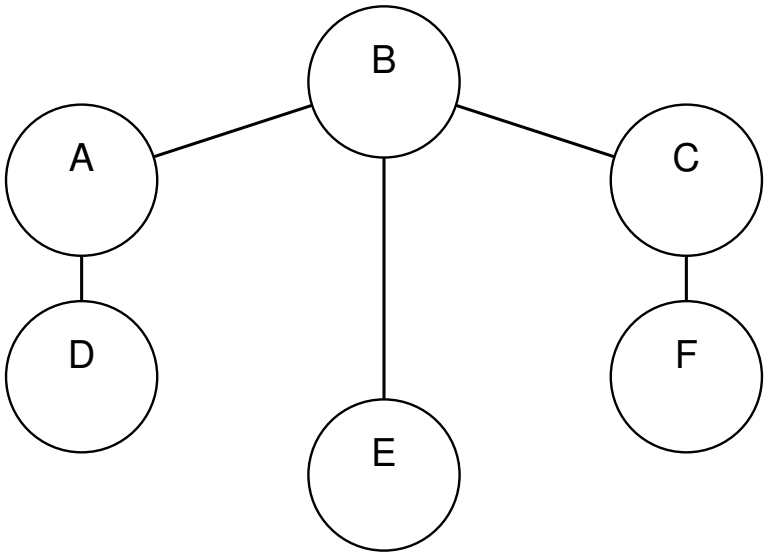


My rule: _____

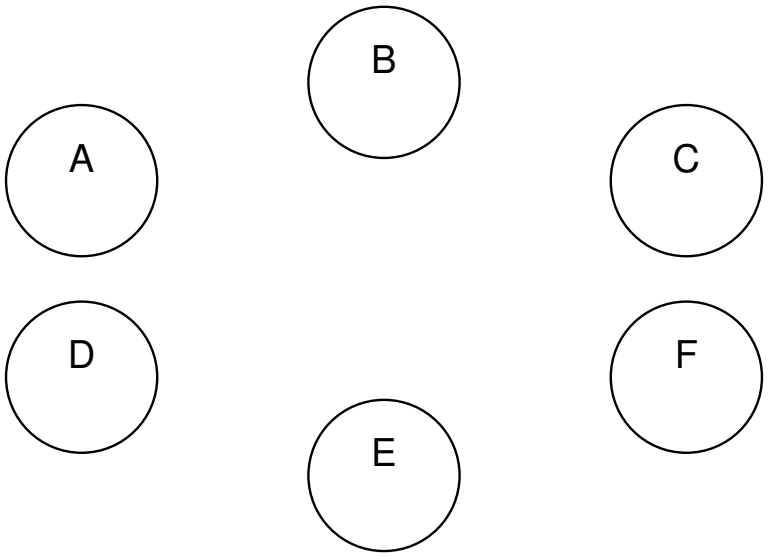
_____

_____

Problem 6: Add the fewest conflict lines you can so each network can no longer fit in two slots. Join only different circles that have no line yet. Show why fewer added lines cannot work.

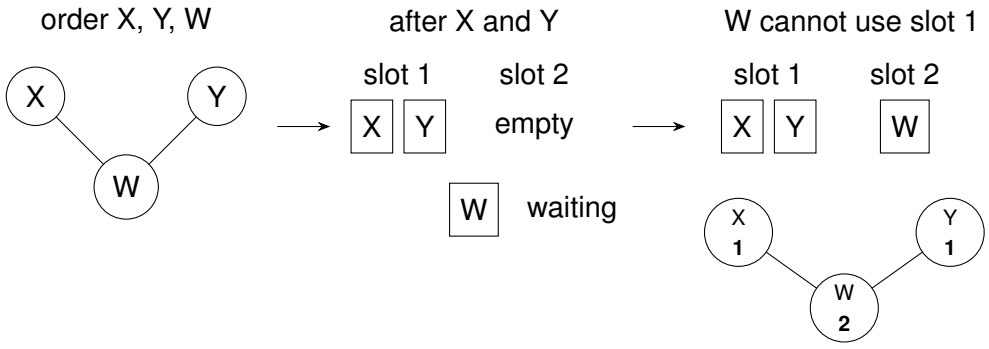


Added lines: _____

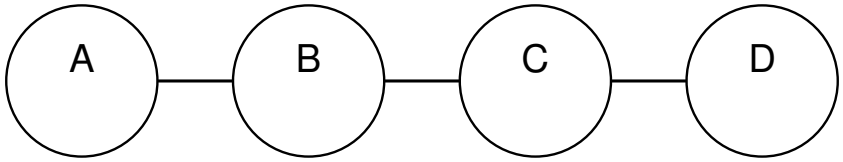


Added lines: _____

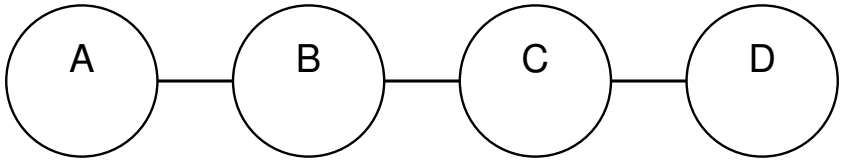
For Problems 7–8, choose an order for the cards. Place each card in the first numbered slot where it conflicts with no card already there. During the order, do not move a placed card.



Problem 7: Find orders that make the first-slot rule use as few slots and as many slots as possible on this network. After an order is finished, you may move cards. Can you use fewer slots?

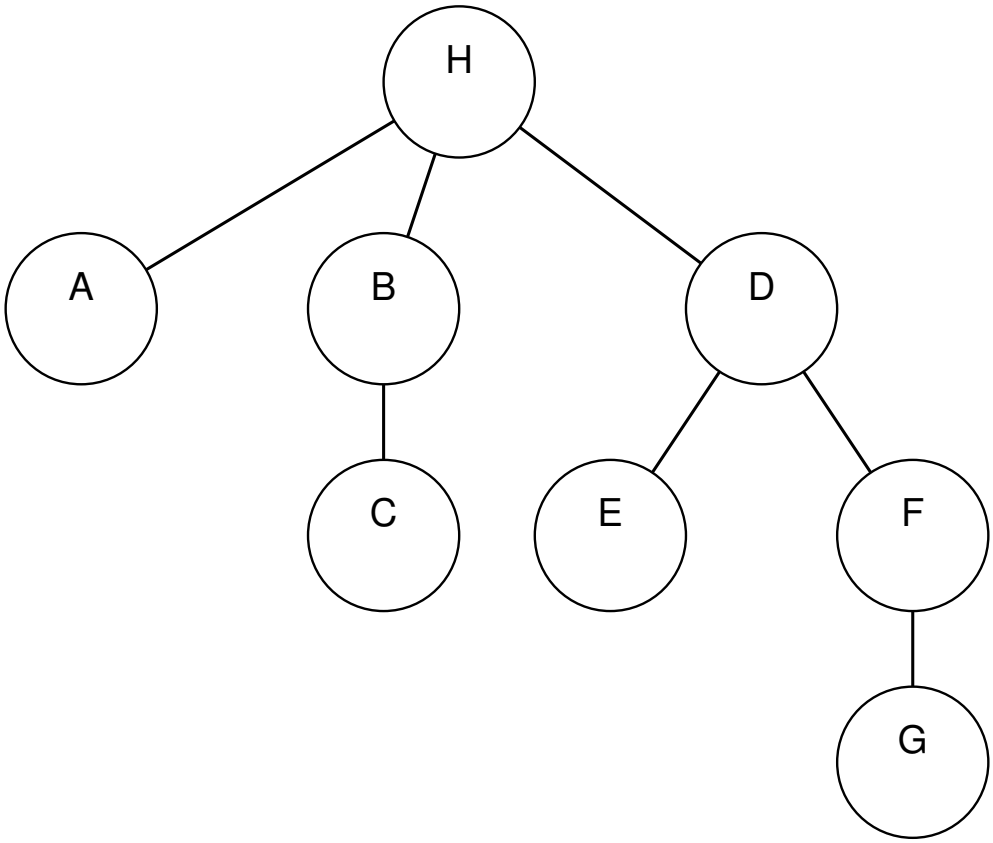


Order: _____



Order: _____

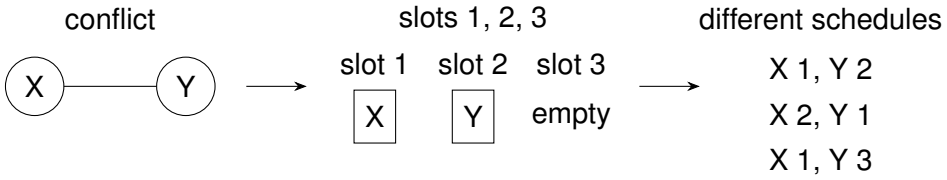
Problem 8: Find an order that makes the first-slot rule use four slots on this network. Find an order that uses two. Could any order use five? Explain.



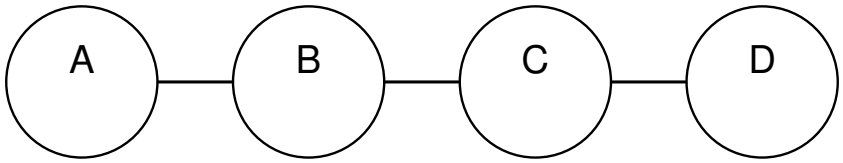
Four-slot order: _____

Two-slot order: _____

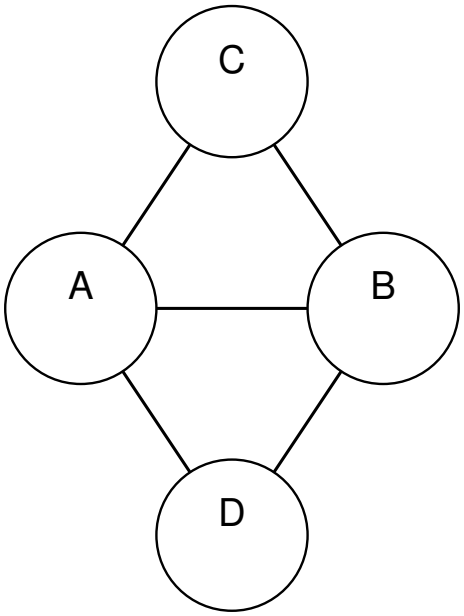
When counting schedules, switching slot numbers counts as a different schedule. Slots may be empty.



Problem 9: How many different schedules does each network have using only slots 1, 2 and 3? How many using slots 1, 2, 3 and 4? Explain how you know your count is complete.



3 slots: _____ 4 slots: _____



3 slots: _____ 4 slots: _____